

Variational Principles for Finite-Slope Pandrosion Flows

Implicit Descent, Local Acceleration, and Lyapunov Structure

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Abstract

This paper proposes a variational interpretation of finite-slope Pandrosion flows. Given an energy or residual functional E , the Pandrosion update is viewed as an implicit descent step driven by a probe-induced finite-slope surrogate. The variational viewpoint explains continuous gates, local acceleration, and conservative behavior far from the root. It also provides a language for proving energy decrease under structural assumptions on the probes, the local finite-slope matrix, and the line-search layer.

1 Finite-slope flow viewpoint

The Pandrosion correction

$$Q_G(a, b)\delta = -G(a)$$

can be interpreted as a discrete flow direction. Rather than approximating a classical derivative, $Q_G(a, b)$ encodes the geometry of a finite path from a to b . The update

$$x_{k+1} = x_k + \lambda_k \delta_k$$

therefore behaves like a sampled flow whose vector field depends on the current probe geometry.

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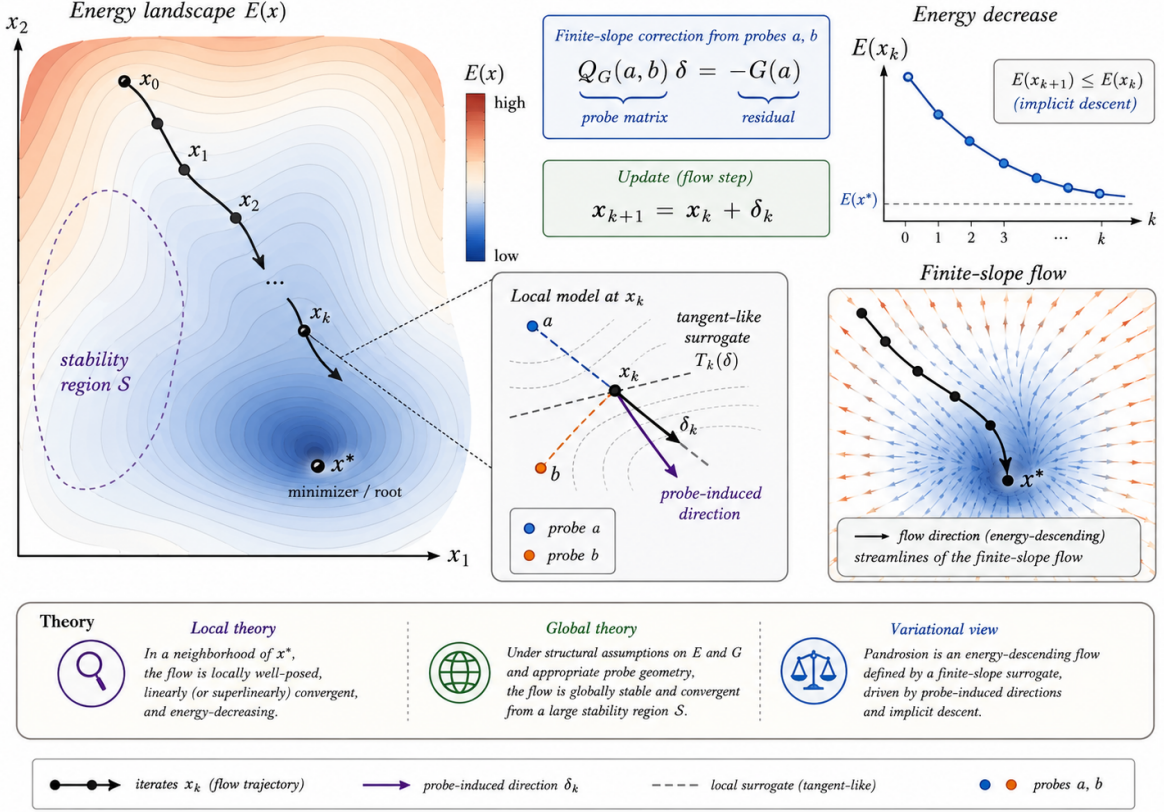


Figure 1: Variational interpretation of finite-slope Pandrosion: probes generate a tangent-like surrogate direction, the line search enforces descent, and the update follows a finite-slope flow toward a root or minimizer.

2 Energy model

Let $E : \mathbb{C}^n \rightarrow \mathbb{R}_+$ be an energy, for example

$$E(x) = \frac{1}{2} \|G(x)\|^2 + \rho \mathcal{L}(x) + \sigma \Pi(x),$$

where $\mathcal{L}$ is a logarithmic scale energy and Π penalizes equal-value or singular probes. A Pandrosion direction is acceptable when it produces an energy decrease after line search:

$$E(x_k + \lambda \delta_k) \leq E(x_k).$$

Definition 2.1 (Finite-slope descent direction). *A direction δ is a finite-slope descent direction at x if it solves $Q_G(x, b)\delta = -G(x)$ for some admissible probe b and if there exists $\lambda > 0$ such that $E(x + \lambda\delta) < E(x)$.*

3 Local descent principle

Proposition 3.1 (Implicit descent condition). *Assume E is locally Lipschitz and that, in a neighborhood of a regular root, the finite-slope matrix $Q_G(x, b)$ is uniformly nonsingular and close to a stable local linearization in the finite-slope sense. Then for x sufficiently close to the root, the Pandrosion direction admits a positive step length λ for which*

$$E(x + \lambda\delta) < E(x).$$

Proof. The finite-slope model gives $G(x) + Q_G(x, b)\delta = 0$. Uniform nonsingularity and finite-slope consistency imply that the residual decreases to first order along sufficiently small multiples of δ . Since E is dominated locally by the residual term up to lower-order scale and penalty terms, a sufficiently small positive step decreases E . $\square$

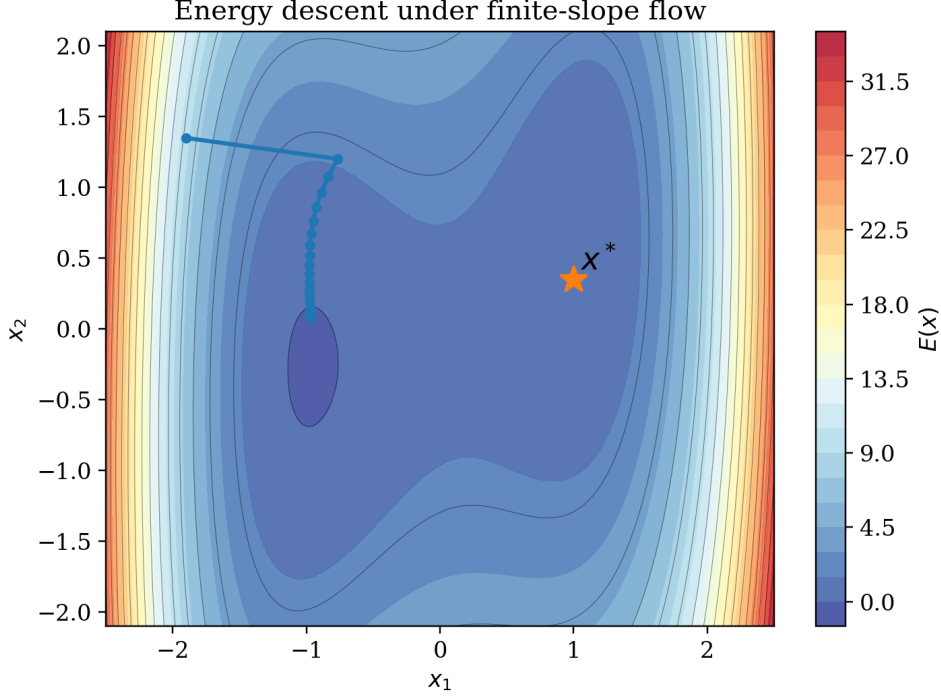


Figure 2: A synthetic energy landscape with a finite-slope trajectory. The path is not a classical gradient method; its local direction is generated from finite-slope probes.

4 Gated local acceleration

The finite-slope Halley correction is best interpreted as a local variational acceleration. Let $\omega(x)$ be a continuous gate depending on residual, condition, and logarithmic scale. The accelerated direction is

$$\delta_\omega = (1 - \omega)\delta_1 + \omega(\delta_1 + \delta_2),$$

where

$$Q_G(a, b)\delta_2 = -\frac{1}{2}H_G(\delta_1, \delta_1)$$

with H_G estimated by symmetric finite-slope probes. Far from the root $\omega \approx 0$; near a stable root $\omega \approx 1$.

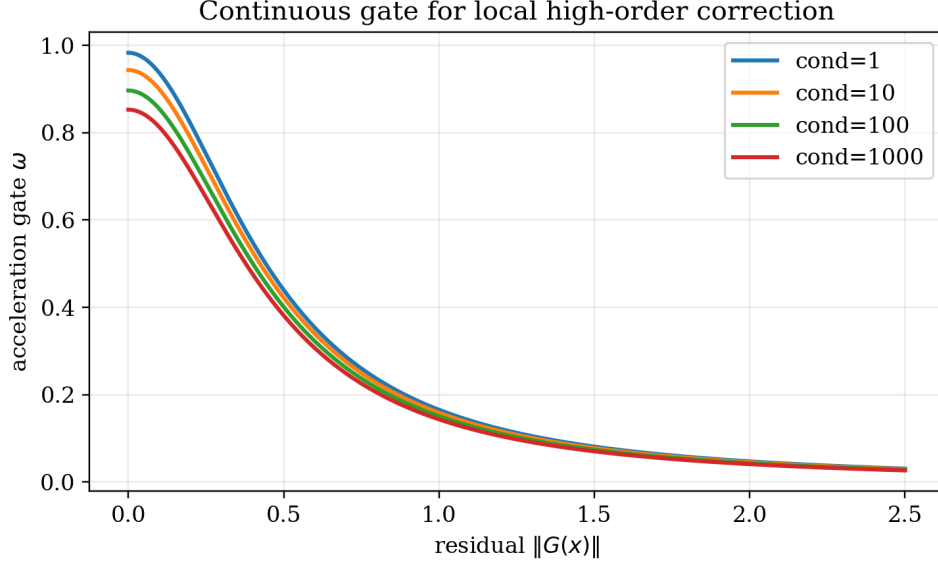


Figure 3: Continuous acceleration gate. High-order corrections are suppressed when the residual is large or the finite-slope matrix is poorly conditioned.

5 Global Lyapunov picture

A global variational theory should use an energy of the form

$$\mathcal{V}(x, i) = E_i(x) + \eta D_i(x),$$

where i is the active chart and D_i is the chart distortion score. Descent can then occur either by changing x inside a chart or by switching charts while reducing D_i .

Theorem 5.1 (Programmatic Lyapunov principle). *Suppose the line-search layer enforces $E_i(x_{k+1}) \leq E_i(x_k)$ within a chart and the atlas layer only accepts chart transitions that decrease D_i up to a bounded tolerance. Then $\mathcal{V}(x_k, i_k)$ is nonincreasing up to the accepted tolerance.*

6 Conclusion

The variational viewpoint provides a clean mathematical bridge between raw Pandrosion, gated finite-slope Halley, projective scaling, and dynamic charts. It also suggests a rigorous future path: prove convergence by constructing explicit Lyapunov energies adapted to finite-slope geometry.

7 Surrogate action functional

The finite-slope direction may be characterized by an implicit surrogate action. Let

$$M_k(\delta) = \frac{1}{2} \|G(x_k) + Q_G(x_k, b_k)\delta\|^2 + \frac{\eta}{2} \|\delta\|^2.$$

The raw Pandrosion correction is the zero-residual minimizer of the first term when Q_G is nonsingular. The regularized problem

$$\delta_k = \arg \min_{\delta} M_k(\delta)$$

provides a variational interpretation even when Q_G is ill-conditioned. In the unregularized case this reduces to the equation $Q_G \delta = -G$.

Proposition 7.1 (Regularized finite-slope direction). *For $\eta > 0$, the surrogate action admits the normal equation*

$$(Q_G^* Q_G + \eta I) \delta = -Q_G^* G(x_k).$$

When Q_G is square and nonsingular and $\eta \rightarrow 0$, the solution converges to the Pandrosion correction.

The practical engines discussed in the Pandrosion program do not need to explicitly solve this normal equation. The proposition is conceptual: it shows that the finite-slope correction can be embedded into an energy-minimization framework.

8 Energy decrease with line search

The line-search layer is the operational mechanism that enforces the variational principle. The algorithm tries a finite grid Λ and chooses

$$\lambda_k \in \arg \min_{\lambda \in \Lambda} E(x_k + \lambda \delta_k).$$

This makes descent robust even when the finite-slope model is only approximate. In this sense, the Pandrosion correction supplies a direction and the line-search layer supplies the variational certificate.

Theorem 8.1 (Discrete descent under accepted steps). *Suppose Λ contains 0 and the algorithm always accepts the grid minimizer of $E(x_k + \lambda \delta_k)$. Then the energy sequence is nonincreasing. If, in addition, the grid contains a positive step that strictly decreases energy whenever x_k is not stationary, then the method cannot remain indefinitely outside the stationary set without decreasing E .*

Proof. Since $0 \in \Lambda$, the minimizer over Λ cannot have energy larger than $E(x_k)$. If a strict descent step exists, the minimizer has strictly smaller energy. Iterating gives monotonicity and excludes stagnation away from the stationary set under the stated assumption. $\square$

9 Finite-slope curvature and second variation

The finite-slope Halley layer estimates a second variation without constructing an analytic Hessian. Along a direction δ one estimates

$$H_G(\delta, \delta) \approx G(x + h\delta) - 2G(x) + G(x - h\delta).$$

The correction

$$Q_G \delta_2 = -\frac{1}{2} H_G(\delta_1, \delta_1)$$

then cancels the leading second-order residual. From the variational viewpoint, this is a local improvement of the surrogate action. It should only be trusted when the first-order finite-slope model is already stable.

10 Noise, probes, and robustness

Probe geometry introduces variability. Different probes generate different surrogates and therefore different descent directions. This is not a defect; it is a source of geometric information. A robust variational Pandrosion engine treats probes as a family of local models and selects the model that minimizes a combination of residual decrease, conditioning, curvature, and chart distortion.

11 Connection with continuous flows

If the step length becomes small and probes vary smoothly with x , the discrete update suggests a continuous flow

$$\dot{x} = -V(x), \quad V(x) = -Q_G(x, b(x))^{-1}G(x).$$

This is not a classical gradient flow unless Q_G is symmetric positive and aligned with a metric. However, it is a natural finite-slope flow. The variational problem is to identify a metric or energy under which V is descent-like.

12 Implementation guidance

The variational interpretation suggests three practical rules:

1. use conservative gates far from a root;
2. line-search every high-order correction;
3. treat chart and probe penalties as part of the energy, not as external heuristics.

These rules explain why the more stable engines in the Pandrosion sequence separate global geometric control from local acceleration.

13 Open problems

The main open problem is to construct explicit energies E for broad classes of polynomial systems such that finite-slope Pandrosion directions are provably descent directions in large regions. A second problem is to characterize when the gated finite-slope Halley correction preserves descent. A third is to connect the dynamic atlas score with a Lyapunov function on an extended state space.

References

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